

Scala For DS

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Practical 19

Combine two plots (E.g , Scatter & line) in a single visualization using Breeze - viz .

Code :

```
import breeze.linalg._
import breeze.plot._

object Scala extends App {

  val x = DenseVector(1.0, 2.5, 3.0, 4.5, 5.0)
  val yscat = DenseVector(5.0, 4.0, 6.0, 8.0, 7.0)
  val yline = DenseVector(1.5, 3.5, 4.5, 5.5, 7.5)

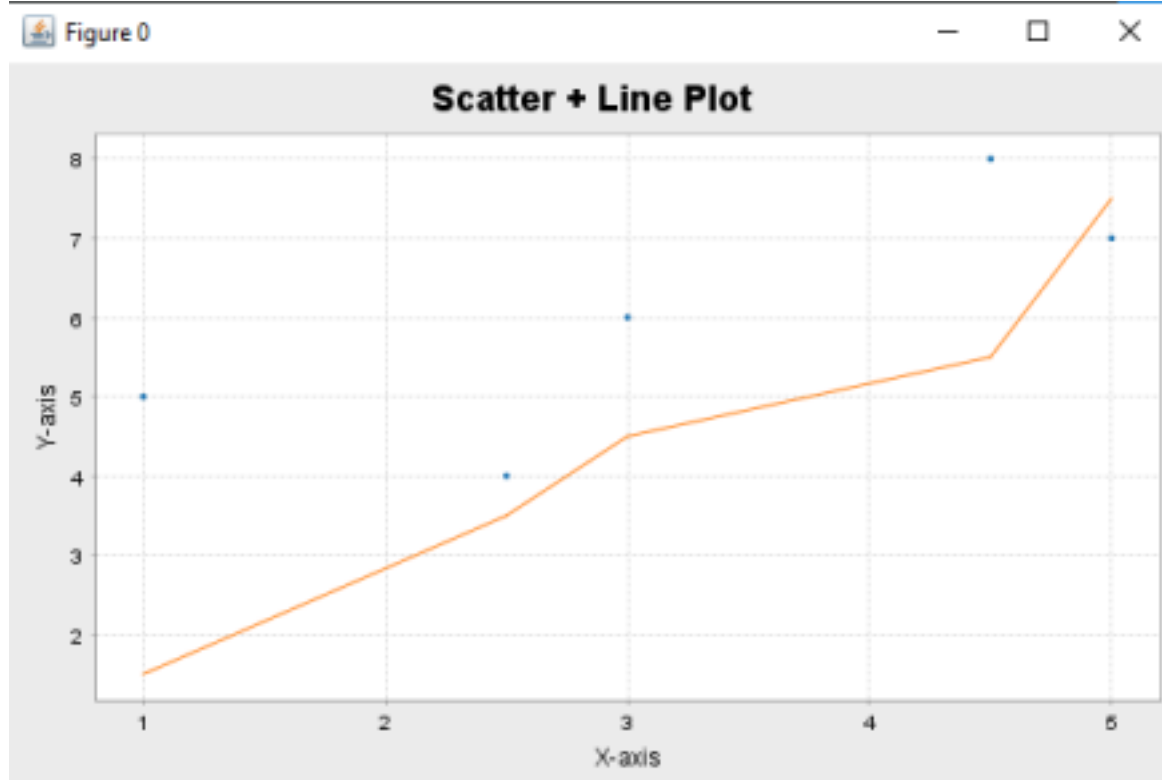
  val fig = Figure()
  val plt = fig.subplot(0)

  plt += plot(x, yscat, '.')
  plt += plot(x, yline, '-')

  plt.title = "Scatter + Line Plot"
  plt.xlabel = "X-axis"
  plt.ylabel = "Y-axis"

  fig.refresh()
}
```

Output :



Practical 20

Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.

Code :

```

import scala.io.Source
object Scala extends App {

    val file = Source.fromResource("iris.csv")

    val data = file.getLines()
        .drop(1)
        .map(line => line.split(",")(0).toDouble)
        .toList

    file.close()

    val count = data.length
    val mean = data.sum / count
    val min = data.min
    val max = data.max

    val variance = data.map(x => math.pow(x - mean, 2)).sum / count
    val std = math.sqrt(variance)

    println(s"Count: $count")
    println(f"Mean: $mean%.2f")
    println(f"Minimum: $min%.2f")
    println(f"Maximum: $max%.2f")
    println(f"Standard Deviation: $std%.2f")
}

```

Output :

```

Count: 150
Mean: 75.50
Minimum: 1.00
Maximum: 150.00
Standard Deviation: 43.30

Process finished with exit code 0
|

```